Projected-Lamb Heat Packing and Sharp Trace Obstructions for Three-Dimensional Navier–Stokes Flow

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Abstract

For the three-dimensional incompressible Navier–Stokes equations, the vorticity stretching and transport terms can be compressed into the curl of the solenoidal Lamb vector $L = \mathbb{P}(u \times \omega)$. This paper studies a signed Littlewood–Paley shell injection built from that representation. The associated vertical heat extension has an unconditional bulk estimate: after normalization by enstrophy, its full heat-height integral belongs to L^1 in physical time for every periodic Leray–Hopf solution on a finite interval. The estimate persists under bounded-overlap spatial localization and yields a conditional local continuation criterion.

The bulk estimate does not recover the bottom trace. A fixed smooth radial tight frame and an explicit global smooth 2D3C Navier–Stokes family along dyadic frequencies $K = 2^j$ give an exact bottom-to-bulk factor $2K^2$. The same family shows that matched partitions do not hide the loss, while a covariant interior-cylinder construction rules out every geometry-only factor $o(r^{-2})$. A second construction uses the finite-dimensional implicit-function theorem for the exact Navier–Stokes flow to prescribe a positive-time full-shell zero with nonzero nonlinear forcing. After an amplitude-frequency scaling, its scale-invariant entry atom is of order λ^{-4}, whereas the bare normalized Leray–Lamb time budget is of order λ^{-6}. Thus no uniform bare payment of such internal entries is possible. An outgoing coarea identity gives an exact scale-matched representation of every finite-order entry, but its zero-level occupation density is not controlled here. None of the results proves global regularity or constructs a singularity. Computer-assisted finite Fourier, spectral-reconstruction, and Galerkin calculations are used as audits and corroboration; no analytic theorem is inferred from floating-point output.

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1 Introduction

Let u be a divergence-free velocity field and $\omega = \operatorname{curl} u$ its vorticity. The obstruction to a direct global enstrophy estimate in three dimensions is the vortex-stretching work $\int S : \omega \otimes \omega$, where $S = (\nabla u + \nabla u^T)/2$. Classical regularity theory controls this work by adding geometric coherence, critical-space smallness, or an integrable strong norm; representative examples include Serrin [9], Constantin and Fefferman [2], and Koch and Tataru [7]. The aim here is narrower. I examine one signed frequency-localized form of the enstrophy injection and determine exactly which part follows from the Leray energy inequality and which part does not.

The starting identity is classical. Modulo a Bernoulli gradient, the Navier–Stokes nonlinearity is the solenoidal Lamb vector

$$L = \mathbb{P}(u \times \omega) = \partial_t u - \nu \Delta u.$$

Solenoidal Lamb decompositions have appeared in fluid mechanics and in mathematical formulations of the curl operator [5, 8, 10]. The contribution of this paper is not that decomposition by itself. It is the specific signed shell quotient below, its energy-level heat packing, and sharp smooth-solution obstructions to recovering its bottom trace.

1.1 Main results

Fix a real-even scalar tight Fourier frame $(T_j)_j$. For heat height $s \geq 0$, set

$$W_{j,s} = e^{s\Delta} T_j \omega, \qquad F_{j,s} = e^{s\Delta} T_j L, \qquad D_{j,s} = \|\nabla W_{j,s}\|_2^2,$$

and define the complete shell injection

$$b_{j,s} = \langle \operatorname{curl} W_{j,s}, F_{j,s} \rangle .$$

The positive shell square function is

$$\Theta_s^2 = \sum_{D_{j,s}>0} \frac{(b_{j,s})_+^2}{D_{j,s}}.$$

The first theorem, proved in Sections 3–4, is

$$\int_0^\infty \Theta_s^2 \, \mathrm{d}s \leq \frac{1}{2} \|L\|_{\dot{H}^{-1}}^2 \leq \frac{1}{2} \|u\|_4^4 . \tag{1}$$

If $Y(t) = \|\omega(t)\|_2^2$ and

$$\mathcal{V}(t) = Y(t)^{-1} \int_0^\infty \Theta_s(t)^2 \, \mathrm{d}s,$$

with the value zero when $Y = 0$, then every periodic Leray–Hopf solution satisfies

$$\int_0^T \mathcal{V}(t) \, \mathrm{d}t \leq \frac{C_S^3}{2\sqrt{2\nu}} \|u_0\|_2^2 \sqrt{T}. \tag{2}$$

This is an unconditional statement at the energy level. Bounded-overlap localization gives the corresponding estimate with an overlap constant and also gives a conditional local continuation criterion.

The bottom value

$$A_{\mathrm{sb},+}(t) = \Theta_0(t)^2 / Y(t)$$

is different. Section 6 constructs one fixed dyadically covariant radial frame and a global smooth 2D3C family for which, at every dyadic frequency $K = 2^j$,

$$A_{\mathrm{sb},+}(0) = 2K^2 \mathcal{V}(0). \tag{3}$$

The identity survives every nonzero nonnegative smooth cutoff. At a parabolic heat height $h = \theta K^{-2}$, the height average is a fixed positive fraction of the bottom value. Matched partitions retain the same scale law. A separate whole-space scaling argument places the obstruction on strictly interior admissible skewed cylinders and rules out a pure geometry-only bottom-from-bulk factor $o(r^{-2})$. The critical Cr^{-2} factor is saturated, not rejected.

Section 7 addresses a temporal objection to a static trace example. Starting from

$$U = (0, \cos x_1, \cos x_2),$$

the exact local Navier–Stokes flow is perturbed in the target shell so that the entire shell vanishes at a prescribed positive time τ, while its nonlinear forcing remains nonzero. The resulting zero is a genuine simple internal positive entry. Choosing a base amplitude $a_\lambda = \lambda^{-2}$ and then applying integer Navier–Stokes dilation gives

$$\frac{\text{internal entry atom}}{\int \|L_\lambda\|^2_{\dot{H}^{-1}} / Y_\lambda \, \mathrm{d}t} = \frac{2\nu}{\sinh(2\nu\tau)}\lambda^2 + o(\lambda^2). \tag{4}$$

The initial kinetic energy and critical $\dot{H}^{1/2}$ norm tend to zero, while the initial enstrophy remains bounded. Thus the failure is not caused by a large energy ledger or an initial-time observation face.

The final positive statement is an exact outgoing-coarea representation of all isolated finite-order entries. Its density has the correct scale, but it concentrates at the distinguished zero level. The Leray inequality does not provide the required uniform occupation estimate in the present argument.

1.2 Relation to nearby work

The heat-extension viewpoint is established in critical Navier–Stokes theory, most notably in the BMO^{-1} framework of Koch and Tataru [7]. The present square function is solution-dependent, signed before squaring, and normalized by enstrophy; it is not the Koch–Tataru norm. Rigorous skewed-cylinder geometry is available from Yang [13] and has been used for derivative estimates by Vasseur and Yang [12]. Those covering theorems do not by themselves recover a boundary value in the independent heat-height variable. Recent caloric trace estimates recover boundary regularity from stronger interior derivative norms [1], rather than from the zero-order bulk in (1).

The closest filtered-vorticity comparison located in the preparation of this draft is Yu [14]. That work controls a positive near-field stretching part and obtains conditional unweighted closure under additional far-field, commutator, and localization summability. The observable and the closure hypotheses differ from those used here. Exact commutator identities also do not imply defect vanishing without an additional compactness or regularity input; this distinction is familiar from Duchon and Robert [3].

The paper makes no priority or literature-exhaustion claim. Its precise scope is summarized in Section 10.

2 Setting and notation

The periodic part of the paper is set on $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ with normalized spatial measure. The velocity has zero mean and satisfies

$$\partial_t u + u \cdot \nabla u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0, \qquad \omega = \operatorname{curl} u. \tag{5}$$

All Fourier multipliers are set to zero on the zero mode. The Leray projection is denoted by $\mathbb{P}$.

A *real-even scalar tight frame* is a countable family of scalar Fourier multipliers $(T_j)_j$, with symbols $m_j(k)$, such that

$$m_j(-k) = m_j(k) \in \mathbb{R}, \qquad \sum_j |m_j(k)|^2 = 1 \quad (k \neq 0). \tag{6}$$

For finite frames all sums are literal. Infinite sums are interpreted by finite truncation followed by monotone or Fatou convergence whenever the summands are nonnegative.

At energy regularity, expressions containing $e^{s\Delta}L$ as an L^2 field are understood for $s > 0$. Their heat-height integrals are defined by finite shell and positive-height truncation followed by Tonelli and Fatou. At $s = 0$, the shell coefficients are used only when $L \in L^2$; this holds, in particular, at almost every H^2 time of a periodic H^1 strong solution. Changing a coefficient on the remaining null set does not affect any continuation hypothesis below.

The rotational form of (5) gives

$$u \times \omega = L + \nabla B, \qquad B = p + \frac{1}{2}|u|^2, \qquad L = \mathbb{P}(u \times \omega) = \partial_t u - \nu\Delta u = -\mathbb{P}\nabla \cdot (u \otimes u). \tag{7}$$

Taking curl yields

$$\partial_t \omega - \nu\Delta\omega = \operatorname{curl} L = S\omega - u \cdot \nabla\omega. \tag{8}$$

For $s \geq 0$, define

$$A_{j,s} = e^{s\Delta}T_j, \quad W_{j,s} = A_{j,s}\omega, \quad F_{j,s} = A_{j,s}L, \quad D_{j,s} = \|\nabla W_{j,s}\|_2^2. \tag{9}$$

Since $W_{j,s}$ is divergence free, $D_{j,s} = \|\operatorname{curl} W_{j,s}\|_2^2$. The heat-extended vorticity equation is

$$(\partial_t - \nu\partial_s)W_{j,s} = \operatorname{curl} F_{j,s}. \tag{10}$$

3 A signed shell continuation coefficient

At heat height s, define the complete shell work

$$b_{j,s} = \langle \operatorname{curl} W_{j,s}, F_{j,s} \rangle. \tag{11}$$

This includes stretching and the transport-filter commutator. Indeed,

$$\begin{aligned} b_{j,s} &= \langle W_{j,s}, A_{j,s}(S\omega - u \cdot \nabla\omega) \rangle \\ &= \langle W_{j,s}, A_{j,s}(S\omega) + [u \cdot \nabla, A_{j,s}]\omega \rangle. \end{aligned} \tag{12}$$

The advective pairing with $W_{j,s}$ vanishes globally. Pressure is the gradient complement in (7) and is therefore not a third independent vorticity work term.

At $s = 0$, write

$$\Theta_0^2 = \sum_{D_{j,0}>0} \frac{(b_{j,0})_+^2}{D_{j,0}}, \qquad A_{\mathrm{sb},+} = \frac{\Theta_0^2}{Y}, \qquad Y = \|\omega\|_2^2, \tag{13}$$

with value zero when $Y = 0$.

Proposition 3.1 (Conditional shell continuation)**.** *Let u be a maximal periodic H^1 strong solution on $[0, T_*)$. If*

$$\int_0^{T_*} A_{\mathrm{sb},+}(t)\,\mathrm{d}t < \infty, \tag{14}$$

then u extends beyond T_*.

Proof. Set $D = \|\nabla\omega\|_2^2$ and $\mathfrak{P} = \int S : \omega\otimes\omega$. Tightness gives $\sum_j D_{j,0} = D$ and $\sum_j b_{j,0} = \mathfrak{P}$. Positive-part subadditivity and weighted Cauchy imply

$$\mathfrak{P}_+ \le \sqrt{D}\,\Theta_0 \le \frac{\nu}{2}D + \frac{1}{2\nu}A_{\mathrm{sb},+}Y.$$

The enstrophy identity therefore yields

$$Y' + \nu D \le \nu^{-1}A_{\mathrm{sb},+}Y.$$

Gronwall and (14) bound Y on $[0,T_*)$ and place D in $L^1(0,T_*)$. The standard strong-solution continuation theorem then extends the solution; see, for example, Fujita and Kato [4], Kato [6], Temam [11]. □

Remark 3.2. Proposition 3.1 is a scale-critical conditional reduction, not an unconditional regularity criterion. The rest of the paper tests whether (14) follows from the energy-controlled heat extension.

4 Unconditional projected-Lamb heat packing

For $s \ge 0$, set

$$\Theta_s^2 = \sum_{D_{j,s}>0} \frac{(b_{j,s})_+^2}{D_{j,s}}. \tag{15}$$

Theorem 4.1 (Global heat-bulk estimate)**.** *At every time at which $u \in H^1(\mathbb{T}^3)$,*

$$\Theta_s^2 \le \left\|e^{s\Delta}L\right\|_2^2 \quad (s>0), \qquad \int_0^\infty \Theta_s^2\,\mathrm{d}s \le \frac{1}{2}\|L\|_{\dot H^{-1}}^2 \le \frac{1}{2}\|u\|_4^4. \tag{16}$$

If $L \in L^2$, the pointwise inequality also holds at $s = 0$. For a periodic Leray–Hopf solution and every finite T, the normalized bulk

$$\mathcal{V}(t) = \frac{1}{Y(t)}\int_0^\infty \Theta_s(t)^2\,\mathrm{d}s \tag{17}$$

satisfies (2).

Proof. Shellwise Cauchy and (11) give

$$\frac{(b_{j,s})_+^2}{D_{j,s}} \le \|F_{j,s}\|_2^2.$$

Summing and using (6) proves the first inequality in (16). Since L has zero mean,

$$\int_0^\infty \left\|e^{s\Delta}L\right\|_2^2\,\mathrm{d}s = \frac{1}{2}\left\|(-\Delta)^{-1/2}L\right\|_2^2.$$

The multiplier $(-\Delta)^{-1/2}\mathbb{P}\nabla\cdot$ has L^2 operator norm at most one from matrix fields with the Frobenius norm to vector fields. Equation (7) therefore gives

$$\|L\|_{\dot H^{-1}} \le \|u\otimes u\|_2 = \|u\|_4^2.$$

This proves (16).

The three-dimensional interpolation and Sobolev inequalities give

$$\|u\|_4^4 \le \|u\|_2 \|u\|_6^3 \le C_S^3 \|u\|_2 Y^{3/2}.$$

Hence

$$\mathcal{V}(t) \le \frac{C_S^3}{2} \|u(t)\|_2 Y(t)^{1/2}.$$

The Leray inequality implies

$$\|u(t)\|_2 \le \|u_0\|_2, \qquad \nu \int_0^T Y(t)\,\mathrm{d}t \le \frac{1}{2}\|u_0\|_2^2.$$

Cauchy in time proves (2).

For weak solutions, the argument is first applied to finite shell and heat-height truncations at almost every H^1 time. Tonelli and Fatou then give the stated interpretation. No solution-dependent test function is inserted into the weak formulation. □

The factorization of the unresolved trace is now explicit. When the denominator is positive, define

$$\Lambda_L^2(t) = \frac{\Theta_0(t)^2}{\displaystyle\int_0^\infty \Theta_s(t)^2\,\mathrm{d}s}; \tag{18}$$

otherwise set $\Lambda_L = 0$. Then

$$A_{\mathrm{sb},+} = \Lambda_L^2 \mathcal{V}. \tag{19}$$

Theorem 4.1 controls $\mathcal{V}$, but does not control Λ_L^2.

5 Bounded-overlap localization

Let $(\chi_{j,Q})_Q$ be a nonnegative smooth partition of unity at the matched radius $r_j \simeq K_j^{-1}$, where T_j is supported in $cK_j \le |k| \le CK_j$. Assume uniform derivative bounds and overlap at most N. Define

$$B_{j,Q}(t,s) = \langle F_{j,s}, \mathrm{curl}(\chi_{j,Q} W_{j,s})\rangle, \qquad d_{j,Q}(t,s) = \|\mathrm{curl}(\chi_{j,Q} W_{j,s})\|_2^2, \tag{20}$$

and set

$$q_{j,Q}(t,s) = \begin{cases} (B_{j,Q}(t,s))_+^2 / d_{j,Q}(t,s), & d_{j,Q} > 0, \\ 0, & d_{j,Q} = 0. \end{cases} \tag{21}$$

The curl in (20) retains the cutoff collar exactly:

$$B_{j,Q} = \int \chi_{j,Q} F_{j,s} \cdot \mathrm{curl}\, W_{j,s} + \int F_{j,s} \cdot (\nabla \chi_{j,Q} \times W_{j,s}).$$

Proposition 5.1 (Local packing and criterion)**.** *Under the preceding assumptions,*

$$\sum_{j,Q} q_{j,Q}(t,s) \le N \left\| e^{s\Delta} L(t) \right\|_2^2. \tag{22}$$

For every $\alpha > 0$,

$$\int_0^\infty s^{\alpha-1} \sum_{j,Q} q_{j,Q}(t,s)\,\mathrm{d}s \le \frac{N\Gamma(\alpha)}{2^\alpha} \left\| (-\Delta)^{-\alpha/2} L(t) \right\|_2^2. \tag{23}$$

At $\alpha = 1$, the normalized left side belongs to $L^1(0,T)$ with the right side of (2) multiplied by N.

Moreover, if

$$A_{\mathrm{loc},+}(t) = \frac{1}{Y(t)} \sum_{j,Q} q_{j,Q}(t,0) \tag{24}$$

is integrable on the maximal strong interval, then the solution continues.

Proof. Cauchy in (20) gives

$$q_{j,Q} \le \left\| 1_{\operatorname{supp} \chi_{j,Q}} F_{j,s} \right\|_2^2 .$$

Bounded overlap and frame tightness prove (22). Spectral integration gives (23); the case $\alpha = 1$ then follows as in Theorem 4.1.

For the criterion, the matched annular Bernstein and collar bounds imply

$$\sum_{Q} d_{j,Q} \le C_\chi \, D_{j,0}.$$

Since $\sum_Q B_{j,Q} = b_{j,0}$, weighted Cauchy gives

$$\frac{(b_{j,0})_+^2}{D_{j,0}} \le C_\chi \sum_{Q} q_{j,Q}(t,0).$$

Thus $A_{\mathrm{sb},+} \le C_\chi A_{\mathrm{loc},+}$, and Proposition 3.1 applies. □

Remark 5.2. The local criterion is not claimed to be weaker than a standard published criterion. Direct Cauchy gives $A_{\mathrm{loc},+} \le N \|L\|_2^2 / Y \le N \|u\|_\infty^2$, so the Serrin endpoint $u \in L_t^2 L_x^\infty$ implies it.

6 Exact critical trace loss

The estimates above integrate in the heat-height variable. This section shows that the missing boundary trace costs two frequency powers even for global smooth solutions.

For $\sigma \in \{-1, 1\}$, $a > 0$, and dyadic integer $K = 2^j$, $j \ge 0$, define

$$u_{\sigma,a,K}(x) = aK \left(0, -2\cos(Kx_1), 2\sigma \sin(Kx_1 + Kx_2) - 2\cos(Kx_2)\right). \tag{25}$$

This field is real, zero mean, divergence free, and independent of x_3. Its horizontal velocity is a decaying two-dimensional shear and its third component solves a linear advection-diffusion equation driven by that shear. It therefore generates a global smooth three-dimensional periodic solution.

Choose one fixed smooth dyadically covariant radial tight frame that first places the radii K and $\sqrt{2}K$ in one parent and then splits them exactly into low and high children. Such a frame is obtained from a smooth square partition in $\log_2 |\xi|$ with a flat top, followed by a fixed sine-cosine refinement. The frame is fixed before K is chosen.

For $\sigma = -1$, the only positive child is the low-radius block. Direct Fourier calculation gives

$$b_{\mathrm{lo}}(s) = 2a^3 K^6 e^{-2K^2 s}, \quad D_{\mathrm{lo}}(s) = 4a^2 K^6 e^{-2K^2 s}, \quad q_{\mathrm{lo}}(s) = a^4 K^6 e^{-2K^2 s}. \tag{26}$$

Theorem 6.1 (Sharp global and localized trace cost)**.** *For the family* (25) *and the fixed frame above,*

$$q_{\mathrm{lo}}(0) = 2K^2 \int_0^\infty q_{\mathrm{lo}}(s)\,\mathrm{d}s, \tag{27}$$

and, with $Y = 8a^2K^4$*,*

$$A_{\mathrm{sb},+}(0) = \frac{a^2K^2}{8}, \qquad \mathcal{V}(0) = \frac{a^2}{16}, \qquad A_{\mathrm{sb},+}(0) = 2K^2\mathcal{V}(0). \tag{28}$$

Let $\phi \geq 0$ *be any nonzero smooth cutoff fixed before the data, and let* q_ϕ *be defined by the complete localized work in* (20)*. Then for every* $h > 0$*,*

$$q_\phi(0) = \frac{2K^2}{1 - e^{-2K^2h}} \int_0^h q_\phi(s)\,\mathrm{d}s. \tag{29}$$

In particular, at $h = \theta K^{-2}$*, the heat-height average is a fixed positive fraction of the bottom value.*

Proof. Equation (27) follows by integrating the last formula in (26); division by Y gives (28).

For the local statement, the low block is

$$W_s = 2aK^2e^{-K^2s}(\sin(Kx_2), 0, \sin(Kx_1)), \quad F_s = -2a^2K^3e^{-K^2s}(0, 0, \cos(Kx_2)).$$

The complete localized work, including the cutoff collar, satisfies

$$B_\phi(s) = \langle F_s, \mathrm{curl}(\phi W_s)\rangle = 4a^3K^6e^{-2K^2s} \int \phi(x)\sin^2(Kx_2)\,\mathrm{d}x > 0.$$

Both F_s and W_s lie on $|k| = K$, so $q_\phi(s) = e^{-2K^2s}q_\phi(0)$. Integration over $[0, h]$ proves (29). □

At the matched height,

$$\frac{h^{-1}\int_0^h q_\phi(s)\,\mathrm{d}s}{q_\phi(0)} = \frac{1 - e^{-2\theta}}{2\theta}. \tag{30}$$

The factor is positive and independent of K.

Matched partitions do not remove the scale. If $(\phi_Q)_Q$ is a standard nonnegative partition at radius $r = \rho/K$ with overlap N and uniform square-derivative bounds, then

$$\frac{a^4K^6e^{-2K^2s}}{2(C_0 + C_1/\rho^2)} \leq \sum_Q q_Q(s) \leq 2Na^4K^6e^{-2K^2s}. \tag{31}$$

Choosing $a = K^{-1}$ keeps $\|u\|_2^2 = 6$. The normalized aggregate bottom remains between positive constants, while its heat bulk is $O(K^{-2})$.

6.1 Interior cylinders

The exact family above is evaluated at an initial trace. The same critical power occurs on strictly interior cylinders, with a narrower conclusion.

For this subsection only, take the covariant one-block tight frame $T_0 = I$. At the center time t_Q of a skewed cylinder Q, let χ_Q be its covariantly transported nonnegative spatial cutoff and set

$$W_s = e^{s\Delta}\omega(t_Q), \qquad F_s = e^{s\Delta}L(t_Q).$$

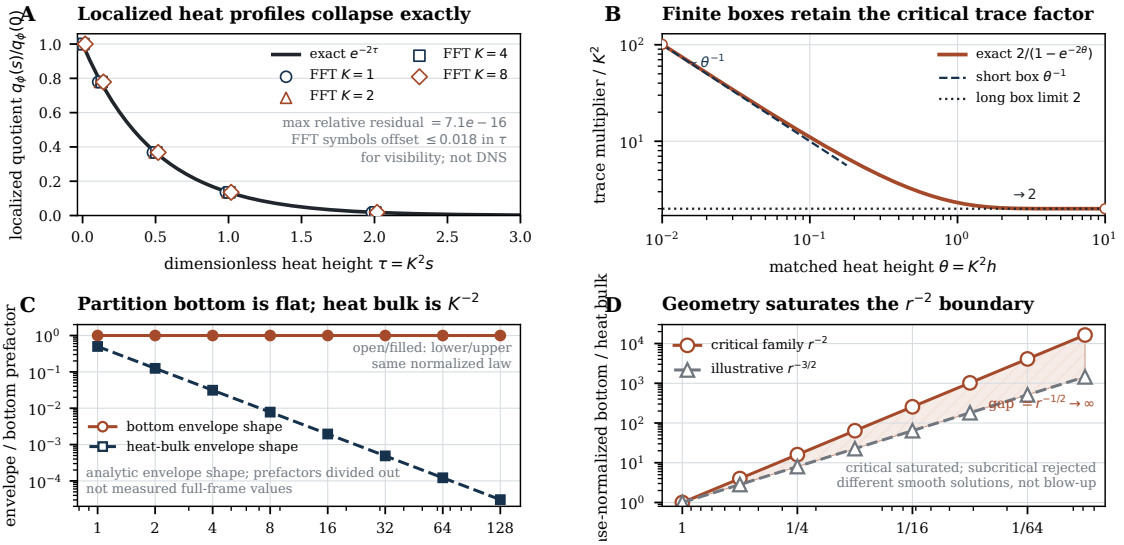


Figure 1: Localized heat packing and the critical trace cost. The exact profile is $q_\phi(s)/q_\phi(0) = e^{-2K^2 s}$. The finite spectral points independently reconstruct the trigonometric identity and are not DNS. Matched partitions preserve the bottom scale while the corresponding bulk decays by K^{-2} at fixed kinetic energy. The last panel shows the covariant r^{-2} interior-cylinder scaling.

Define

$$B_Q(s) = \langle F_s, \operatorname{curl}(\chi_Q W_s)\rangle\,, \qquad d_Q(s) = \|\operatorname{curl}(\chi_Q W_s)\|_2^2\,, \tag{32}$$

$$q_Q(s) = \begin{cases} (B_Q(s))_+^2 / d_Q(s), & d_Q(s) > 0, \\ 0, & d_Q(s) = 0. \end{cases}$$

With $Y_Q = \|\omega(t_Q)\|_2^2$, put

$$A_Q = \frac{q_Q(0)}{Y_Q}, \qquad \mathcal{V}_{Q,h} = \frac{1}{Y_Q}\int_0^h q_Q(s)\,\mathrm{d}s. \tag{33}$$

Theorem 6.2 (Geometry-only subcritical trace no-go)**.** *There is a scale-covariant family of local-in-time smooth whole-space Navier–Stokes solutions, each defined on an interval containing a strictly interior admissible skewed cylinder Q_r, for which*

$$\frac{A_{Q_r}}{\mathcal{V}_{Q_r,\theta r^2}} = c_* r^{-2}, \qquad c_* > 0. \tag{34}$$

Consequently, an estimate

$$A_Q \le c(r)\mathcal{V}_{Q,\theta r^2}$$

that is required for all smooth solutions and all cylinders in this covariant one-block class, with constants depending only on the uniform geometry and admissibility parameters, cannot have $c(r) = o(r^{-2})$.

Proof. Let

$$u_{\mathrm{aff}} = (x_1, -x_2 - x_3, 0), \qquad \omega_{\mathrm{aff}} = (1, 0, 0).$$

Put $A_0 = -\frac{1}{3}x \times u_{\mathrm{aff}}$, and choose $\eta \in C_c^\infty(\mathbb{R}^3)$ equal to one on a ball. Then $u_0 = \operatorname{curl}(\eta A_0)$ is smooth, compactly supported, and divergence free. Because u_{aff} is linear, homogeneous, and divergence free, $\operatorname{curl} A_0 = u_{\mathrm{aff}}$, so $u_0 = u_{\mathrm{aff}}$ on the smaller affine core.

In the affine core, $\operatorname{curl} L = (\omega\cdot\nabla)u - (u\cdot\nabla)\omega = (1,0,0)$. Choose a nonnegative cutoff χ supported in the core with $\nabla\chi \times \omega_{\mathrm{aff}} \not\equiv 0$. At the initial trace,

$$\int \chi\, \omega_{\mathrm{aff}} \cdot \operatorname{curl} L \,\mathrm{d}x > 0, \qquad \|\operatorname{curl}(\chi\omega_{\mathrm{aff}})\|_2 > 0.$$

Multiplying the datum by a sufficiently small fixed positive amplitude preserves both inequalities. Smooth local existence and continuity preserve them at a small positive center time. Sufficiently small skewed cylinders are admissible in the sense of Yang [13], because their admissibility quantity is bounded by $r^2 \sup \|\nabla u\|_\infty$.

Fix one such base cylinder with radius r_0. Continuity in s gives $q_Q(s) > 0$ on some interval $[0, h_0]$; set $\theta = h_0/r_0^2$. Under the exact scaling

$$u_\lambda(t,x) = \lambda u(\lambda^2 t, \lambda x), \quad r_\lambda = \lambda^{-1} r_0, \quad h_\lambda = \lambda^{-2} h_0,$$

and covariant scaling of the cutoff, mollifier, trajectory, and heat height,

$$B_{Q_\lambda} = \lambda^3 B_Q, \quad d_{Q_\lambda} = \lambda^3 d_Q, \quad q_{Q_\lambda} = \lambda^3 q_Q, \quad Y_{Q_\lambda} = \lambda Y_Q.$$

Consequently $A_{Q_\lambda} = \lambda^2 A_Q$, while $\mathcal{V}_{Q_\lambda,h_\lambda} = \mathcal{V}_{Q,h_0}$. Since both base quantities are positive, this is exactly (34), with $c_* = r_0^2 A_Q/\mathcal{V}_{Q,h_0} > 0$. □

Theorem 6.2 does not reject a critical Cr^{-2} trace inequality, a theorem with additional collar or solution-dependent terms, or a statement on cylinders that stay a fixed positive physical time away from the initial face as $r \to 0$.

7 A genuine positive-time internal entry

The trace obstructions above leave open whether positive-time dynamics could prevent the relevant shell observable from entering through zero. This section constructs such an entry for the exact smooth Navier–Stokes flow.

Let

$$U(x) = (0, \cos x_1, \cos x_2), \qquad \Omega = \operatorname{curl} U = (-\sin x_2, 0, -\sin x_1). \tag{35}$$

The projected Lamb field is

$$F_* = \mathbb{P}(U \times \Omega) = (0, 0, \cos x_1 \sin x_2). \tag{36}$$

It is supported on the four modes $(\pm 1, \pm 1, 0)$, hence on $|k|^2 = 2$, while U lies on $|k|^2 = 1$. Normalized torus averages give

$$\|\Omega\|_2^2 = 1, \quad \|F_*\|_2^2 = \frac{1}{4}, \quad \|\operatorname{curl} F_*\|_2^2 = \frac{1}{2}, \quad -\Delta F_* = 2F_*. \tag{37}$$

Let E_* be the finite-dimensional real divergence-free velocity space on the target modes and let P_* be the corresponding projection. Fix an integer $m \geq 3$. Standard periodic local theory gives a smooth fixed-time solution map S_t from a neighborhood of zero in the divergence-free, mean-zero H^m space to H^m, on a common interval; see Fujita and Kato [4], Temam [11]. Its derivative is

$$DS_\tau(0)h = e^{\nu\tau\Delta} h. \tag{38}$$

Theorem 7.1 (Smooth internal positive entry)**.** *For every sufficiently small fixed $\tau > 0$, there is $a_0 > 0$ and a smooth map $z : (-a_0, a_0) \to E_*$ such that*

$$z(a) = -a^2 \tau F_* + O(a^3), \tag{39}$$

and, for every $0 < |a| < a_0$, the solution with initial datum $u_0^a = aU + z(a)$ is smooth on $[0, 2\tau]$ and has a simple positive target-shell entry at the strictly internal time $t = \tau$. Its leading entry coefficient is

$$A_+(a) = \frac{a^2 e^{-2\nu\tau}}{4} + O(a^3), \qquad A_-(a) = 0. \tag{40}$$

At least one cell in every nonnegative covering partition also has a simple positive entry at that time.

Proof. Define

$$\Phi(a, z) = P_* S_\tau(aU + z).$$

At the origin, $D_z\Phi(0, 0) = e^{-2\nu\tau} I_{E_*}$, an invertible finite matrix. The ordinary implicit-function theorem gives a smooth $z(a)$ with $\Phi(a, z(a)) = 0$. The quadratic Duhamel term on the target shell is

$$e^{\nu(\tau-r)\Delta} P_* \mathbb{P}\big((e^{\nu r\Delta} U) \times \operatorname{curl}(e^{\nu r\Delta} U)\big) = e^{-2\nu\tau} F_*.$$

Integration in r gives (39).

At time τ, the target-shell velocity and vorticity vanish, whereas the projected Lamb forcing satisfies

$$P_* L(u^a(\tau)) = a^2 e^{-2\nu\tau} F_* + O(a^3) \neq 0.$$

Let $W_* = T_*\omega$ and $C_* = \operatorname{curl} W_*$. From the filtered vorticity equation and $W_*(\tau) = 0$,

$$C_{*,t}(\tau) = -\Delta P_* L(u^a(\tau)).$$

The target forcing is divergence free, annular, and nonzero. Therefore

$$\langle P_* L(u^a(\tau)), C_{*,t}(\tau)\rangle = \|\nabla P_* L(u^a(\tau))\|_2^2 > 0.$$

The zero is simple and enters from the positive side. Equations (37) and the leading low-shell enstrophy give (40).

For a nonnegative partition $(\chi_Q)_Q$, the full-shell root gives

$$\langle P_* L, C_{Q,t}\rangle = \int \chi_Q \, |\operatorname{curl} P_* L|^2 \, \mathrm{d}x.$$

The sum over Q is positive, so at least one cell has a positive simple entry. □

Lemma 7.2 (Global positive entries are simple)**.** *Every finite-order positive entry of a global nonzero annular shell is simple.*

Proof. At a global-shell zero, $C_j = \operatorname{curl} W_j = 0$. Since W_j is divergence free, annular, and zero mean, this implies $W_j = 0$. Hence $C_{j,t} = -\Delta F_j$. If the zero had order greater than one, then $F_j = 0$, contradicting positivity of the entry pairing. Finally, $\langle F_j, C_{j,t}\rangle = \|\nabla F_j\|_2^2 > 0$. □

8 No uniform bare payment

For a finite-order entry at t_β, write

$$C_\beta(t_\beta + s) = c_\beta s^{m_\beta} + O_{L^2}(|s|^{m_\beta+1}), \qquad c_\beta \neq 0,$$

and define

$$A_{\beta,+} = \frac{\left\langle F_{j(\beta)}(t_\beta), c_\beta / \|c_\beta\|_2 \right\rangle_+^2}{Y(t_\beta)}, \qquad a_\beta = \kappa_{j(\beta)}^{-2} A_{\beta,+}. \tag{41}$$

Here κ_j is the nominal shell frequency. In the base construction of Theorem 7.1, the target modes have Fourier radius $|k| = \sqrt{2}$, and the shell convention is $\kappa_* = 1$; after the integer dilation λ, the corresponding nominal frequency is $\kappa_{*,\lambda} = \lambda$. With this convention the atom a_β is scale invariant.

For the family in Theorem 7.1,

$$R(a) := \int_0^{2\tau} \frac{\|L^a(t)\|_{\dot H^{-1}}^2}{Y^a(t)}\,\mathrm{d}t = \frac{a^2(1 - e^{-4\nu\tau})}{16\nu} + O(a^3). \tag{42}$$

Choose $a_\lambda = \lambda^{-2}$ and apply the integer periodic scaling

$$u_\lambda(x,t) = \lambda u^{a_\lambda}(\lambda x, \lambda^2 t). \tag{43}$$

Theorem 8.1 (Internal scaling no-go)**.** *Along* (43)*, the constructed internal entry and the bare budget satisfy*

$$a_{\beta,\lambda} = \frac{e^{-2\nu\tau}}{4\lambda^4} + O(\lambda^{-6}), \tag{44}$$

$$R_\lambda = \frac{1 - e^{-4\nu\tau}}{16\nu\lambda^6} + O(\lambda^{-8}), \tag{45}$$

and therefore

$$\frac{a_{\beta,\lambda}}{R_\lambda} = \frac{2\nu}{\sinh(2\nu\tau)}\lambda^2 + o(\lambda^2). \tag{46}$$

There is no constant C, uniform along this family, for which the sum of internal entry atoms is bounded by CR_λ.

Moreover,

$$\|u_\lambda(0)\|_2^2 = O(\lambda^{-2}), \qquad \|u_\lambda(0)\|_{\dot H^{1/2}}^2 = O(\lambda^{-1}), \qquad \|\omega_\lambda(0)\|_2^2 = 1 + o(1). \tag{47}$$

Proof. The base entry is given by (40). The covariant dilation moves it to time τ/λ^2 and preserves the scale-zero atom; the amplitude choice contributes $a_\lambda^2 = \lambda^{-4}$, proving (44). The base budget (42) acquires the additional time-scaling factor λ^{-2}, which gives (45). Their quotient is (46).

On the normalized torus, the dilation multiplies squared L^2, $\dot H^{1/2}$, and $\dot H^1$ norms by λ^2, λ^3, and λ^4, respectively. Combining these factors with $a_\lambda^2 = \lambda^{-4}$ proves (47). □

Theorem 8.1 excludes only the bare right side in (42) with a constant uniform along the covariant family. It does not exclude a fixed-frequency theorem, a constant depending on high Sobolev norms, or an estimate with an added atomic, variation, strong-Lamb, or material-derivative charge.

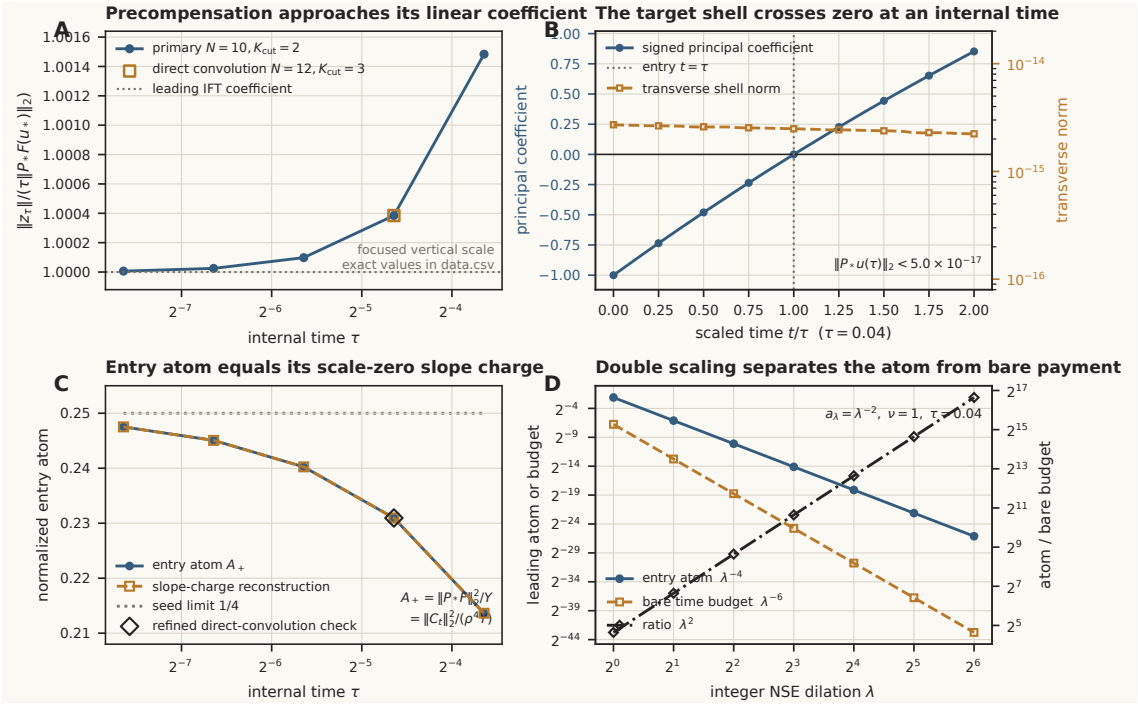


Figure 2: Finite Fourier–Galerkin corroboration of the internal-entry geometry. Shooting a finite-dimensional target-shell correction makes the target shell vanish at $t = \tau$; the analytic continuum result is Theorem 7.1, not the time stepper. The last panel displays the exact leading powers in Theorem 8.1.

9 Outgoing occupation at the zero level

The bare budget has the wrong scale for an atomic entry. A zero-level occupation density has the correct scale and represents the atom exactly.

Let I be a compact classical interval whose endpoints are not zeros. For a finite family Λ, assume that every zero of every C_α, $\alpha \in \Lambda$, is isolated and of finite order. Put

$$r_\alpha(t) = \|C_\alpha(t)\|_2\,, \qquad \xi_\alpha(t) = \frac{C_\alpha(t)}{r_\alpha(t)} \quad (r_\alpha > 0), \tag{48}$$

and

$$q_\alpha(t) = \frac{\left\langle F_{j(\alpha)}(t), \xi_\alpha(t)\right\rangle_+^2}{Y(t)}. \tag{49}$$

Choose $\rho \in C_c^\infty((0,1))$, $\rho \ge 0$, $\int_0^1 \rho = 1$, and set $\rho_\delta(s) = \delta^{-1}\rho(s/\delta)$.

Theorem 9.1 (Outgoing-coarea identity)**.** *Under the preceding assumptions,*

$$\begin{aligned}&\sum_{\alpha\in\Lambda} \sum_{\substack{t_*\in I\\ C_\alpha(t_*)=0}} \kappa_{j(\alpha)}^{-2} A_{\alpha,+}(t_*) \\ &\quad = \lim_{\delta\downarrow 0} \sum_{\alpha\in\Lambda} \kappa_{j(\alpha)}^{-2} \int_I q_\alpha(t)\rho_\delta(r_\alpha(t))\,(r_{\alpha,t}(t))_+ \,\mathrm{d}t.\end{aligned} \tag{50}$$

The identity includes both odd crossings and even touches.

Proof. Near a zero of order m,

$$C(t_* + s) = cs^m + O_{L^2}(|s|^{m+1}), \quad r(t_* + s) = \|c\|_2 \, s^m(1 + O(s)), \quad \xi(t_* + s) \longrightarrow c/\|c\|_2$$

from the right. The factor $(r_t)_+$ removes the incoming left side. On the right, the substitution $y = r(t)$ gives

$$\int q(t)\rho_\delta(r(t))\,(r_t(t))_+ \, \mathrm{d}t \longrightarrow q(t_*+) = A_+(t_*).$$

The argument is unchanged for an even-order touch. Summing finitely many disjoint zero neighborhoods proves (50). □

If $C_t - \nu\Delta C = G$, then wherever $r > 0$,

$$r_t = \langle \xi, G \rangle - \nu \frac{\|\nabla C\|_2^2}{r}. \tag{51}$$

The density in (50) has scale exponent two before time integration and therefore total scale zero. This repairs the dimensional mismatch of the bare budget only at the level of representation. The bound $(r_t)_+ \le \|G\|_2$ still leaves $\rho_\delta(r) \sim \delta^{-1}$, and no uniform zero-level occupation estimate follows from an ordinary $L_t^p G$ bound.

For comparison, a fixed smooth divergence-free packet ψ has an amplitude-weighted excursion bound that is paid by the Leray ledger. If $b_\psi(t) = \langle u(t), \psi \rangle$, then on $I = [s, t]$,

$$V_I^+(b_\psi) \le \nu \, \|\nabla\psi\|_2 \, (|I| \, D_I)^{1/2} + \|\nabla\psi\|_\infty \, E_I \, |I| \,, \tag{52}$$

where $E_I = \operatorname{ess\,sup}_I \|u\|_2^2$ and $D_I = \int_I \|\nabla u\|_2^2$. This controls entries that reach a fixed amplitude, not arbitrarily shallow zero entries.

10 Discussion and exact claim boundary

The results isolate a precise separation.

1. The projected-Lamb shell injection has a complete signed identity and a scale-critical conditional continuation coefficient.
2. Leray energy unconditionally controls the normalized vertical heat bulk, globally and under bounded-overlap localization.
3. The bottom trace is not a free consequence of that bulk. Exact global smooth solutions saturate the full K^2 trace cost, and covariant interior cylinders saturate r^{-2}.
4. Positive-time internal shell entries genuinely occur in smooth Navier–Stokes dynamics. The bare normalized $\dot{H}^{-1}$-Lamb time integral cannot uniformly pay their scale-invariant atoms.
5. Outgoing occupation gives an exact scale-matched representation, but its summed zero-level mass is an open estimate.

The paper does not prove that $A_{\mathrm{sb},+} \in L_t^1$ for every Leray–Hopf solution, does not prove an occupation or recurrence bound, and does not extend the finite-order entry ledger through an unknown singular time. It neither proves global regularity nor constructs finite-time breakdown. The remaining question is not another algebraic decomposition: it is whether the exact Navier–Stokes coupling supplies a scale-zero occupation, jet, or amplitude-thresholded charge that can be summed from an a priori budget.

Computer-assisted verification and reproducibility

Computer assistance was used in four bounded roles: finite Fourier algebra checks, independent spectral and real-space reconstructions, quadrature and scaling-table audits, and figure generation. These calculations test the finite-frame identities and the numerical points reported in the figures. The continuum arguments remain analytic: they use the projected-Lamb identity, heat-semigroup estimates, the local solution map, the implicit-function theorem, and the Leray energy inequality. No theorem in this paper depends on a floating-point output being treated as a proof.

The exact Fourier identities, independent reconstructions, figure data, commands, environments, and checksums used here are archived in the immutable public snapshot at commit `89b2ba3cc808`. The localized-trace certificate and the internal-entry certificate contain the exact producers, independent programs, environments, outputs, and checksum manifests. The time-stepping experiment shown in Figure 2 is a finite Fourier–Galerkin corroboration, not DNS and not a continuum error certificate.

A Finite Fourier ledger for the six-mode trace family

For completeness, the vorticity of (25) is

$$\omega_{\sigma,a,K} = 2aK^2\left(\sigma\cos(Kx_1+Kx_2)+\sin(Kx_2), -\sigma\cos(Kx_1+Kx_2), \sin(Kx_1)\right).$$

Normalized averages give

$$\|u\|_2^2 = 6a^2K^2, \qquad \|\omega\|_2^2 = 8a^2K^4, \qquad \|\nabla\omega\|_2^2 = 12a^2K^6. \tag{53}$$

Resolving the complete nonlinear enstrophy work by output radius gives

output radius	stretching	transport	combined
K	0	$2\sigma a^3K^6$	$-2\sigma a^3K^6$
$\sqrt{2}K$	$2\sigma a^3K^6$	$-2\sigma a^3K^6$	$4\sigma a^3K^6$

Thus stretching and the transport-filter commutator must remain together. For a radial block with multiplier values α at K and β at $\sqrt{2}K$, the complete work is

$$b_T = 2\sigma a^3K^6(2\beta^2-\alpha^2),$$

while at heat height s, with $\tau_h = K^2s$,

$$b_T(s) = 2\sigma a^3K^6\left(2\beta^2e^{-4\tau_h}-\alpha^2e^{-2\tau_h}\right).$$

The fixed radial refinement used in Section 6 sets one child to $(\alpha,\beta)=(1,0)$ and the other to $(0,1)$, which yields (26) for $\sigma=-1$.

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